

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP20017
Subject Title	Mechanics and Robotic Motion
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	AP10005 Requires students to be able to calculate simple derivatives, integrals and matrix representation and operation.
Objectives	The objectives of this subject are to (i) further develop the fundamental theories of mechanics taught in AP10005 Physics I or AP10008 University Physics I; (ii) introduce topics at intermediate level of mechanics which are considered to be the foundation for pursuing higher degree studies or performing related engineering analyses; and (iii) introduce elemental concepts of robotics from the mechanical perspective, including geometry and kinematics of robot systems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) select an appropriate coordinate system and apply Newton's laws of motion (with the concepts of linear/angular momentum and energy, and derivatives and integrals) to solve problems of motion of a particle; (b) solve problems of damped and forced oscillation of a particle by solving equation of motions (differential equations); (c) solve problems of motion of a many-particle system, with emphasis put on the steady mass flow system and the variable mass system; (d) solve problems of kinematics and kinetics of rigid body with both translational and rotational motion, using the concepts of Newton's laws, energy and angular momentum. (e) understand the essential concepts of robotics including spatial descriptions of robot components, transformation matrix between different space, forward kinematics, inverse kinematics, and trajectory generation.
Subject Synopsis/ Indicative Syllabus	Kinematics of particles: Cartesian; polar; and cylindrical coordinate systems. Kinetics of a particle: Newton's laws of motion; linear and angular momentum; potential energy; kinetic energy; work-energy theorem; Vibration: free vibration without damping; damped and forced harmonic oscillation. Many-particle system: Center of mass; linear and angular momentum; kinetic energy and work; steady mass flow systems; variable mass systems; motion of rigid body. Robotics: spatial descriptions of robot components; transformation matrix between different space; forward kinematics; inverse kinematics; trajectory generation.

Teaching/Learning Methodology	Lecture: Delivery of lectures interactively to enable students to participate actively in acquiring knowledge, and to raise questions and discuss for clarifying their doubts generated in their learning process. Examples in relation to real life and engineering practices are given to enable the student to alert applications of the subject contents to real life problems. Tutorial: To be run after introducing the essential concepts for the students to consolidate the contents of lectures. Students are properly guided to participate actively in solving problems, raising questions and discussion. e-learning: Electronic means and multimedia technologies would be adopted for presentations of lectures; communication between students and lecturer; delivery of handouts, homework and notices etc. in order to enhance the effectiveness of teaching and learning processes. Laboratory: Two experiments will be conducted, covering selected topics highlighted in intended learning outcomes. Students will work in groups and conduct the experiments under the guidance of the teaching staff. They are required to analyze their experimental results and complete lab reports.																																	
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1" data-bbox="520 860 1474 1308"> <thead> <tr> <th data-bbox="520 860 820 1061" rowspan="2">Specific assessment methods/tasks</th><th data-bbox="820 860 975 1061" rowspan="2">% weighting</th><th colspan="5" data-bbox="975 860 1474 994">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th data-bbox="975 994 1062 1061">a</th><th data-bbox="1062 994 1150 1061">b</th><th data-bbox="1150 994 1254 1061">c</th><th data-bbox="1254 994 1358 1061">d</th><th data-bbox="1358 994 1474 1061">e</th></tr> </thead> <tbody> <tr> <td data-bbox="520 1061 820 1167">1. Continuous assessment</td><td data-bbox="820 1061 975 1167">40</td><td data-bbox="975 1061 1062 1167">✓</td><td data-bbox="1062 1061 1150 1167">✓</td><td data-bbox="1150 1061 1254 1167">✓</td><td data-bbox="1254 1061 1358 1167">✓</td><td data-bbox="1358 1061 1474 1167">✓</td></tr> <tr> <td data-bbox="520 1167 820 1234">2. Examination</td><td data-bbox="820 1167 975 1234">60</td><td data-bbox="975 1167 1062 1234">✓</td><td data-bbox="1062 1167 1150 1234">✓</td><td data-bbox="1150 1167 1254 1234">✓</td><td data-bbox="1254 1167 1358 1234">✓</td><td data-bbox="1358 1167 1474 1234">✓</td></tr> <tr> <td data-bbox="520 1234 820 1308">Total</td><td data-bbox="820 1234 975 1308">100 %</td><td colspan="5" data-bbox="975 1234 1474 1308"></td></tr> </tbody> </table> Continuous assessment: The continuous assessment includes assignments, quizzes and test(s) which aim at checking the progress of students study throughout the course, assisting them in fulfilling the learning outcomes. Assignments in general include end-of-chapter problems, which are used to reinforce and assess the concepts and skills acquired by the students; and to let them know the level of understanding that they are expected to reach. At least one test would be administered during the course of the subject as a means of timely checking of learning progress by referring to the intended outcomes, and as means of checking how effective the students digest and consolidate the materials taught in the class. Examination: This is a major assessment component of the subject. It would be a closed-book examination. Complicated formulas would be given to avoid rote memory, such that the emphasis of assessment would be put on testing the understanding, analysis and problem solving ability of the students.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Continuous assessment	40	✓	✓	✓	✓	✓	2. Examination	60	✓	✓	✓	✓	✓	Total	100 %					
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	Other student study effort:	
	▪ Self-study	75 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	Meriam, J.L., Kraige, L.G., “Engineering Mechanics, Dynamics”	